

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Problem-Solving and Data Analysis

05

10 answers · Rationale-rich · Teach from doc

Q1**B****B.** $\frac{5}{12}$

Choice B is correct. Since the participant will be selected at random, the probability of selecting a participant from group A, given that the participant is at least 10 years of age, is equal to the number of participants from group A who are at least 10 years of age divided by the total number of participants who are at least 10 years of age. Based on the table, in group A, there are 17 participants who are 10–19 years of age and 8 participants who are 20+ years of age. Therefore, there are a total of $17 + 8$, or 25, participants in group A who are at least 10 years of age. Based on the table, of the total number of participants, there are 30 participants who are 10–19 years of age and 30 participants who are 20+ years of age. Therefore, a total of $30 + 30$, or 60, of the participants are at least 10 years of age. Thus, the probability of selecting a participant from group A, given that the participant is at least 10 years of age, is $\frac{25}{60}$, or $\frac{5}{12}$.

Choice A is incorrect. This is the number of participants from group A who are at least 10 years of age divided by the total number of participants, rather than divided by the number of participants who are at least 10 years of age.

Choice C is incorrect. This is the probability of randomly selecting a participant from group A, given that the participant is 10–19 years of age, rather than given that the participant is at least 10 years of age.

Choice D is incorrect. This is the probability of randomly selecting a participant who is at least 10 years of age, given that the participant is in group A.

Q2**D****D.** 4,941%

Choice D is correct. It's given that a is 2,241% of the sum of b and c . This can be represented by the equation $a = \frac{2,241}{100}b + c$, or $a = 22.41b + c$. It's also given that b is 83% of c . This can be represented by the equation $b = \frac{83}{100}c$, or $b = 0.83c$. Dividing both sides of this equation by 0.83 yields $\frac{b}{0.83} = c$. Substituting $\frac{b}{0.83}$ for c in the equation $a = 22.41b + c$ yields $a = 22.41b + \frac{b}{0.83}$, or $a = 22.41\frac{1.83b}{0.83}$, which is equivalent to $a = \frac{41.0103b}{0.83}$, or $a = 49.41b$. This equation is equivalent to $a = \frac{4,941}{100}b$; therefore, a is 4,941% of b .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3**41**

The correct answer is 41. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of 75 objects and has a mean of 25 meters. This can be represented by the equation $\frac{x}{75} = 25$, where x represents the sum of the heights of the objects, in meters, in data set A. Multiplying both sides of this equation by 75 yields $x = 7525$, or $x = 1,875$ meters. Therefore, the sum of the heights of the objects in data set A is 1,875 meters. It's also given that data set B consists of the heights of 50 objects and has a mean of 65 meters. This can be represented by the equation $\frac{y}{50} = 65$, where y represents the sum of the heights of the objects, in meters, in data set B. Multiplying both sides of this equation by 50 yields $y = 5065$, or $y = 3,250$ meters. Therefore, the sum of the heights of the objects in data set B is 3,250 meters. Since it's given that data set C consists of the heights of the 125 objects from data sets A and B, it follows that the mean of data set C is the sum of the heights of the objects, in meters, in data sets A and B divided by the number of objects represented in data sets A and B, or $\frac{1,875 + 3,250}{125}$, which is equivalent to 41 meters. Therefore, the mean, in meters, of data set C is 41.

Q4**4176**

The correct answer is 4,176. It's given that the side length of the larger square is 3 times the side length of the smaller square. This means that the area of the larger square is 3^2 , or 9, times the area of the smaller square. If the area of the smaller square is represented by x , then the area of the larger square can be represented by $9x$. Therefore, the flat surface of the two adjacent squares has a total area of $x + 9x$, or $10x$. It's given that an electric field with strength 29.00 volts per meter passes uniformly through this surface and the total electric flux of the electric field through this surface is 4,640 volts · meters. Since it's given that the electric flux is the product of the electric field's strength and the area of the surface, the equation $29.0010x = 4,640$, or $290x = 4,640$, can be used to represent this situation. Dividing each side of this equation by 290 yields $x = 16$. Substituting 16 for x in the expression for the area of the larger square, $9x$, yields 916, or 144, square meters. Since the area of the larger square is 144 square meters, the electric flux, in volts · meters, of the electric field through the larger square can be determined by multiplying the area of the larger square by the strength of the electric field. Thus, the electric flux is 144 square meters $\frac{29.00 \text{ volts}}{\text{meter}}$, or 4,176 volts · meters.

Q5**A**

- A. Each year between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

Choice A is correct. The given equation is in slope-intercept form, or $y = mx + b$, where m is the value of the slope of the line of best fit. Therefore, the slope of the line of best fit is 0.096. From the definition of slope, it follows that an increase of 1 in the x -value corresponds to an increase of 0.096 in the y -value. Therefore, the line of best fit predicts that for each year between 1940 and 2010, the minimum wage will increase by 0.096 dollar per hour.

Choice B is incorrect and may result from using the y -coordinate of the y -intercept as the average increase, instead of the slope. Choice C is incorrect and may result from using the 10-year increments given on the x -axis to incorrectly interpret the slope of the line of best fit. Choice D is incorrect and may result from using the y -coordinate of the y -intercept as the average increase, instead of the slope, and from using the 10-year increments given on the x -axis to incorrectly interpret the slope of the line of best fit.

Q6

42

The correct answer is 42. It's given that in Nigeria on September 30, 2015, the probability of selecting an MTN subscriber from all Internet subscribers is 0.43, that there were p million, or $p(1,000,000)$, MTN subscribers, and that there were 97 million, or 97,000,000, Internet subscribers. The probability of selecting an MTN subscriber from all Internet subscribers can be found by dividing the number of MTN subscribers by the total number of Internet subscribers.

Therefore, the equation $\frac{p(1,000,000)}{97,000,000} = 0.43$ can be used to solve for p . Dividing 1,000,000 from the numerator and denominator of the expression on the left-hand side yields $\frac{p}{97} = 0.43$. Multiplying both sides of this equation by 97 yields $p = (0.43)(97) = 41.71$, which, to the nearest integer, is 42.

Q7**A****A.** 124.28%

Choice A is correct. It's given that the value of the comic book increased by 167% from the end of 2011 to the end of 2012. Therefore, if the value of the comic book at the end of 2011 was x dollars, then the value, in dollars, of the comic book at the end of 2012 was $x + \frac{167}{100}x$, which can be rewritten as $1x + 1.67x$, or $2.67x$. It's also given that the value of the comic book decreased by 16% from the end of 2012 to the end of 2013. Therefore, the value, in dollars, of the comic book at the end of 2013 was $2.67x - 2.67x\frac{16}{100}$, which can be rewritten as $2.67x - 2.67x0.16$, or $2.2428x$. Thus, if the value of the comic book at the end of 2011 was x dollars, and the value of the comic book at the end of 2013 was $2.2428x$ dollars, then from the end of 2011 to the end of 2013, the value of the comic book increased by $2.2428x - 1x$, or $1.2428x$, dollars. Therefore, the increase in the value of the comic book from the end of 2011 to the end of 2013 is equal to 1.2428 times the value of the comic book at the end of 2011. It follows that from the end of 2011 to the end of 2013, the net percentage increase in the value of the comic book was 1.2428100%, or 124.28%.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the difference between the net percentage increase in the value of the comic book from the end of 2011 to the end of 2012 and the net percentage decrease in the value of the comic book from the end of 2012 to the end of 2013, not the net percentage increase in the value of the comic book from the end of 2011 to the end of 2013.

Choice D is incorrect. This is the net percentage increase in the value of the comic book from the end of 2011 to the end of 2013, if the value of the comic book increased by 167% from the end of 2011 to the end of 2012 and then increased, not decreased, by 16% from the end of 2012 to the end of 2013.

Q8**B****B.** II only

Choice B is correct. The mean of a data set is the sum of the values divided by the number of values. It's given that a new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. The histogram shows that all the values in data set A are greater than 18. Since the additional number of points, 18, is less than any of the values in the original data set, the mean number of points per player for data set B is less than the mean number of points per player for data set A. The median of a data set with 50 values is between the 25th and 26th values when the values are listed in ascending order. The median of a data set with 51 values is the 26th value when the values are listed in ascending order. Therefore, the median of data set B is the 26th value in data set B, which is the 25th value in data set A. The histogram shows that the 25th and 26th values in data set A are both between 50 and 60 points. Therefore, the 25th and 26th values in data set A could be equal, in which case the median of data set A would be equal to both the 25th and 26th values in data set A. Then, the median number of points per player for data set A would be equal to the median number of points per player for data set B. Therefore, only statement II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**.0625**

Accept also: 1/16

The correct answer is .0625 . It's given that $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$. The equation $\frac{4a}{b} = 6.7$ can be rewritten as $4\frac{a}{b} = 6.7$. Dividing both sides of this equation by 4 yields $\frac{a}{b} = 1.675$. The equation $\frac{a}{bn} = 26.8$ can be rewritten as $\frac{a}{b} \frac{1}{n} = 26.8$. Substituting 1.675 for $\frac{a}{b}$ in this equation yields $1.675 \frac{1}{n} = 26.8$, or $\frac{1.675}{n} = 26.8$. Multiplying both sides of this equation by n yields $1.675 = 26.8n$. Dividing both sides of this equation by 26.8 yields $n = 0.0625$. Therefore, the value of n is 0.0625 . Note that .0625, 0.062, 0.063, and 1/16 are examples of ways to enter a correct answer.

Q10

285

The correct answer is 285. The number of people predicted to visit the beach each day is represented by the y-values of the line of best fit, and the average temperature, in degrees Celsius ($^{\circ}\text{C}$), is represented by the x-values. Since the slope of the line of best fit is approximately 57, the y-value, or the number of people predicted to visit the beach each day, increases by 57 for every x-value increase of 1, or every 1°C increase in average temperature. Therefore, an increase of 5°C in average temperature corresponds to a y-value increase of $57(5) = 285$ additional people per day predicted to visit the beach.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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